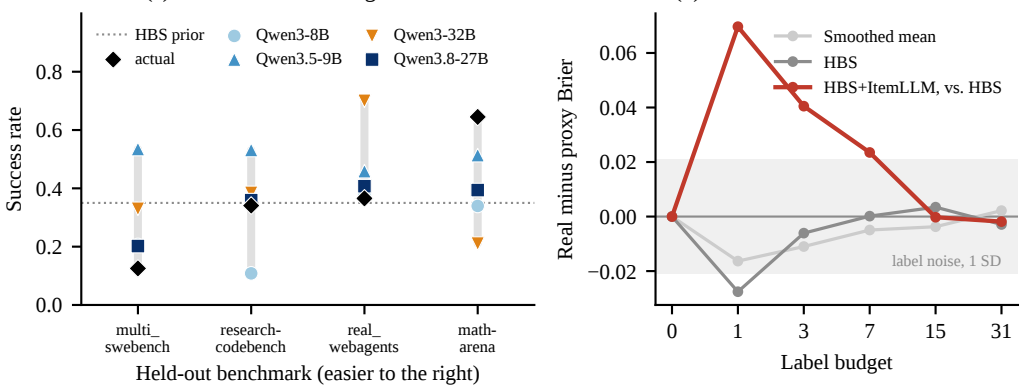
(a) Anchored LLM ratings vs actual
HBS prior
actual
Qwen3-8B
Qwen3.5-9B
Qwen3-32B
Qwen3.8-27B
Success rate
0.0
0.2
0.4
0.6
0.8
multi_ swebench
research- codebench
real_ webagents
math- arena
Held-out benchmark (easier to the right)
(b) HBS+ItemLLM on its real draw
Smoothed mean
HBS
HBS+ItemLLM, vs. HBS
Real minus proxy Brier
0.06
0.04
0.02
0.00
−0.02
label noise, 1 SD
0
1
3
7
15
31
Label budget